

Steganography

What is Steganography?

- **Steganography, in simple words, means hiding one piece of data within another. For example, consider the sentence:**
 - **“Since everyone can read, encoding text in neutral sentences is doubtfully effective.”**
- **Since Everyone Can Read, Encoding Text In Neutral Sentences Is Doubtfully Effective**
- **The hiding info: "SECRET INSIDE"**

History of Steganography

- The ancient pharaohs hieroglyph,
- Herodotus, 440 BC, two incidents,
- Demaratus had sent a warning about an attack to Greece by writing it on the wooden backing of a wax tablet before applying its beeswax surfaces,
- Ancient Chinese wrote messages on fine silk, crunched into a tiny ball, covered by wax, then swallowed by the messenger,
- Special "inks" were important steganographic tools during Second World War and after

Physical Techniques

- Hidden Messages within wax tablets,
- Hidden messages on messenger's body,
- Hidden messages on paper written in secret inks,
- Messages written in Morse code on knitting yarn, then knitted into a piece of clothing worn by a courier,
- Messages written on envelopes in the area covered by postage stamps.

Digital Techniques

- Concealing messages within the lowest bits of noisy images or sound files,
- Modifying the echo of a sound file (Echo Steganography),
- Including data in ignored sections of a file, such as after the logical end of the carrier file

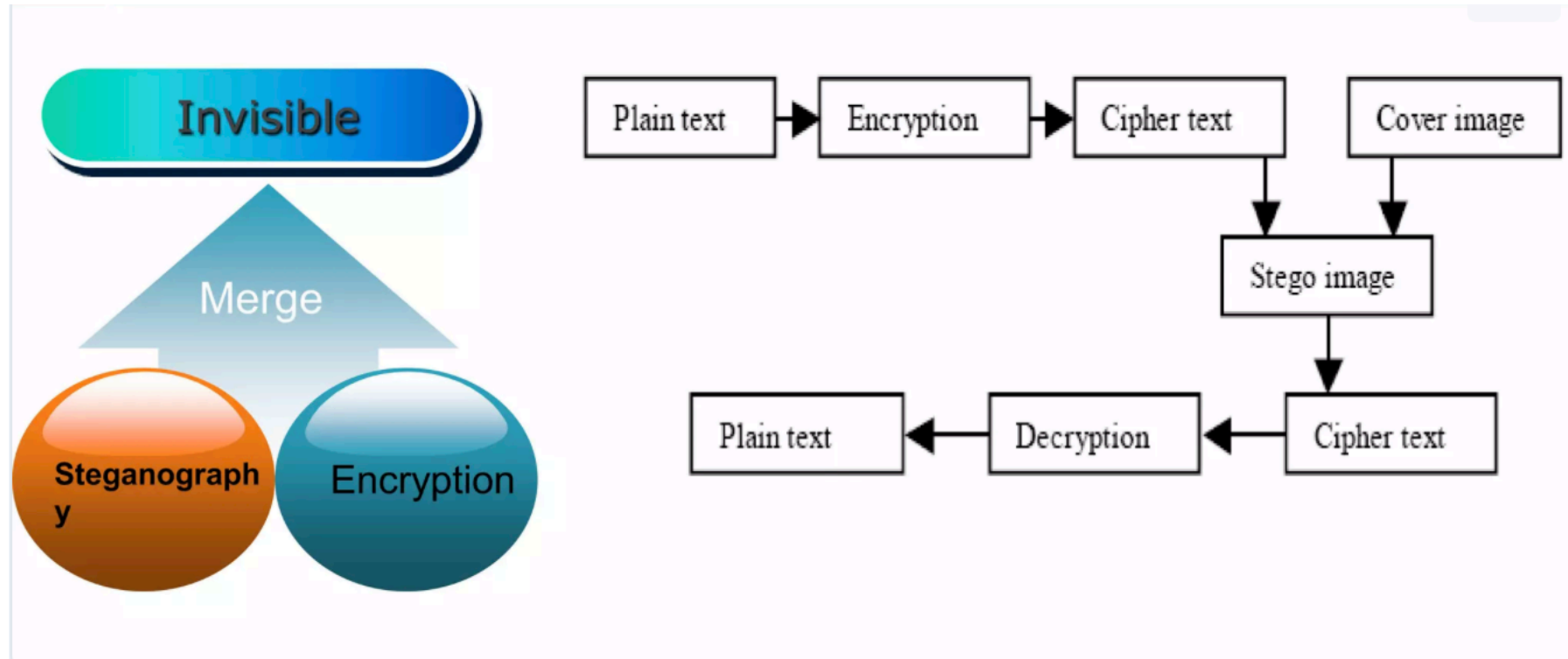
Capabilities Comparison

Secret Communication Technique	Confidentiality	Integrity	Availability
Encryption	Yes	No	Yes
Digital Signature	No	Yes	No
Steganography	Yes / No	Yes / No	Yes

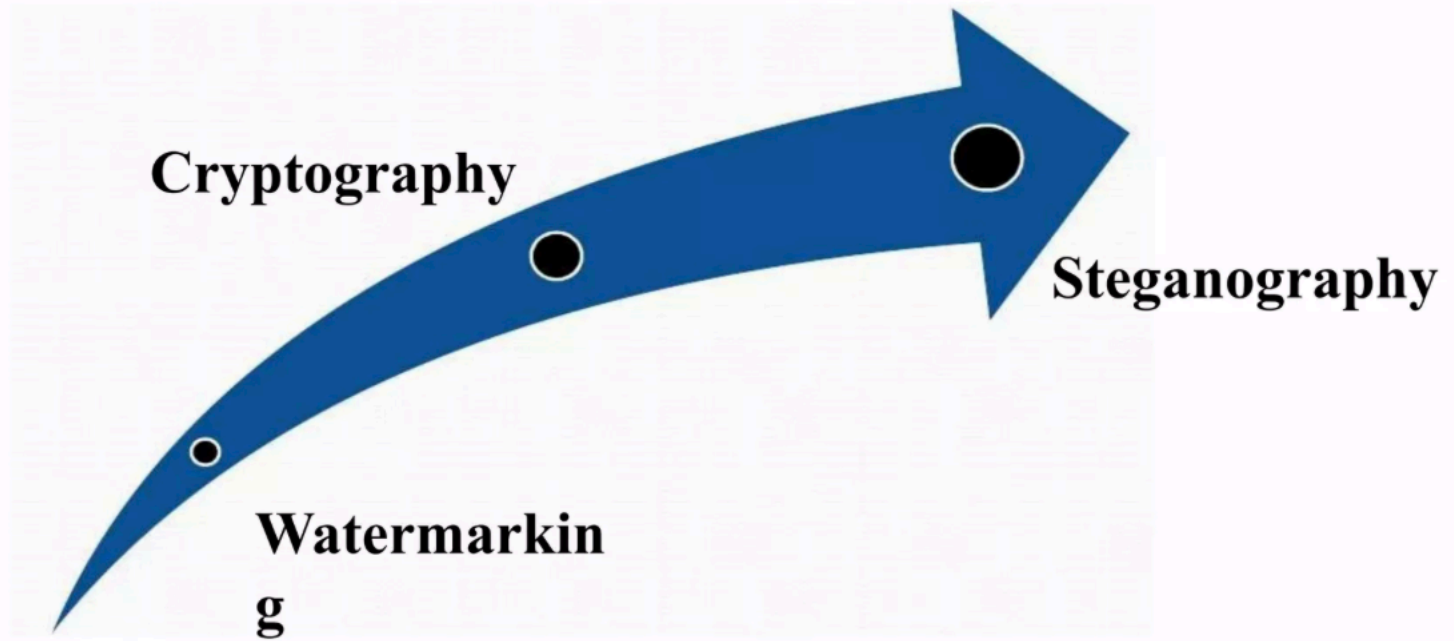
Steganography Versus Cryptography

Steganography	Cryptography
Unknown message passing	Known message passing
Steganography prevents discovery of the very existence of communication	Encryption prevents an unauthorized party from discovering the contents of a communication
Little known technology	Common technology
Technology still being develop for certain formats	Most of algorithm known by all
Once detected, message is known	Strong current algorithm are resistant to attacks, larger expensive computing power is required for cracking
Steganography does not alter the structure of the secret message	Cryptography alter the structure of the secret message

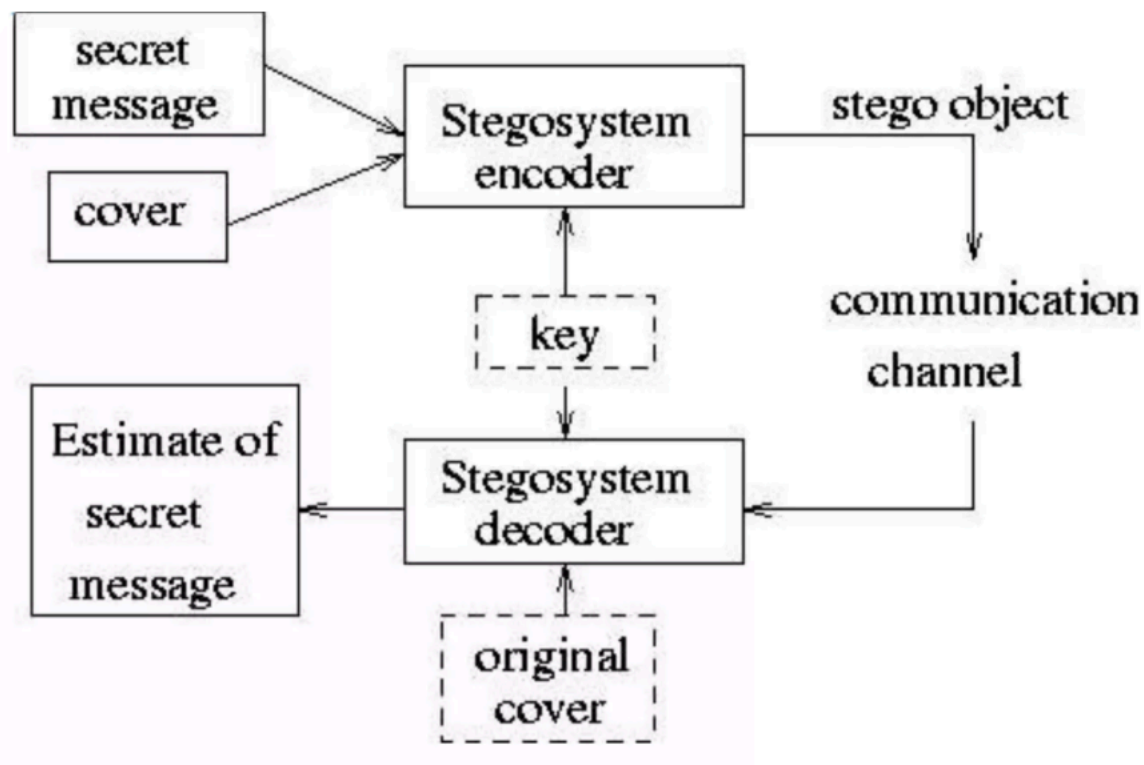
Combined Crypto-Steganography



Evolution



Basic Steganography Model



Steganography Terms

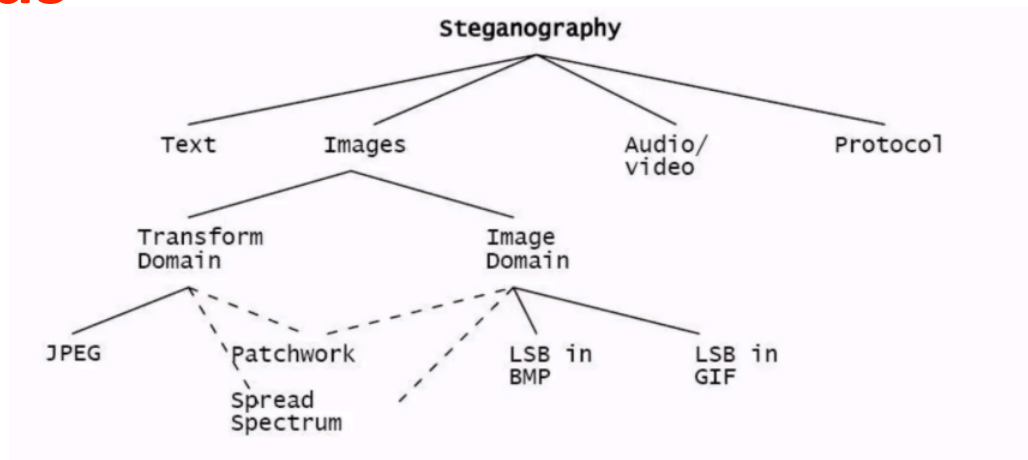
- Carrier or Cover File - the Original message or file in which hidden information will be stored inside it,
- Stego-Medium - The medium in which the information is hidden,
- Embedded or Payload - The information which is to be hidden or concealed,
- Steganalysis - The process of detecting hidden information inside a file

Types of Stegosystems

- **Three basic types of Stegosystems :**
 - **Pure stegosystems - no key is used,**
 - **Secret-key stegosystems - secret key is used,**
 - **Public-key stegosystems - public key is used**

Image Steganography

- Using image files as hosts for steganographic messages takes advantage of the limited capabilities of the human visual system,
- Some of the most common methods for embedding messages in image files can be categorized into two main groups, image domain methods and transform domain methods



LSB [Least Significant Bit] Method

- Is a common, simple approach to embedding information in a cover image,
- The least significant bit (in other words, the 8th bit) of some or all of the bytes inside an image is changed to a bit of the secret,
- When using a 24-bit image, a bit of each of red, green and blue colors are used, since they are represented by a byte,
- In other words, one can store 3 bits in each pixel, An 800×600 pixel image, can thus store a total amount of 1,440,000 bits or 180,000 bytes of embedded data,
- LSB uses BMP images (lossless compression)

Example of the LSB Method

- A grid for 3 pixels of a 24-bit image can be as follows:

(00101101 00011100 11011100)

(10100110 11000100 00001100)

(11010010 10101101 01100011)

- When the number 200, which binary representation is **11001000**, is embedded into the least significant bits of this part of the image, the resulting grid is as follows:

(0010110**1** 0001110**1** 11011100)

(1010011**0** 1100010**1** 00001100)

(1101001**0** 1010110**0** 01100011)

Steganography Tools

1. Anubis
2. BMP Secrets
3. DarkCryptTC
4. OpenPuff
5. OpenStego
6. StegFS
7. StegoShare

Other tools: https://en.wikipedia.org/wiki/Steganography_tools

Best steganography tools: <http://www.topbestalternatives.com/best-steganography-software/>

StegoShare



StegoShare allows embedding of large files into multiple images

OpenPuff



Free tool that implements three layers of hidden data

